

Lab 5

1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?
 - a. Converting raw audio into spectrograms makes the important sound patterns easier for the model to recognize words more easily. Spectrograms show which frequencies are present and when they happen.
2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?
 - a. The representative dataset gives the model a small test set to run using real-world data before it gets converted. It provides the range of lowest and highest number for the model so it knows how to scale/quantize the number accurately and make sure the model stays small at the same time.
3. What does `audio_processor.get_data()` do? Refer to `input_data.py`.
 - a. It builds one batch of training/validation/test inputs from the datasets.
4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are `g_model` and `g_model_len`, and why do they need to be copied correctly into the Arduino source file?
 - a. `g_model` is a C/C++ byte array (unsigned char) that contains the raw, serialized binary data of your quantized .tflite model.
 - b. `g_model_len` is an integer that specifies the exact size of this array in bytes
 - c. The microcontroller has no file system to "load" from the tflite file. It needs to be flashed directly into the microcontroller's program memory (usually Flash memory) right alongside your compiled C/C++ code, rather than taking up its precious and limited working memory (RAM).
5. Please include a screenshot showing the correct identification of at least one keyword, either "YES" or "NO", as displayed in the Arduino IDE Serial Monitor after deploying your KWS model on the Arduino Nano 33 BLE.

Heard unknown (211) @173520ms

Heard unknown (213) @175584ms

Heard yes (201) @177456ms

Heard unknown (202) @178896ms

Heard unknown (204) @183568ms

Heard no (201) @185184ms

Heard unknown (217) @187344ms